

Graphing Quadratic Expressions

Example Equations of Linear Graphs (covered previously)

$$y = 2x - 1, y + 2 = 3(x - 1), -5x + 2y = 7$$

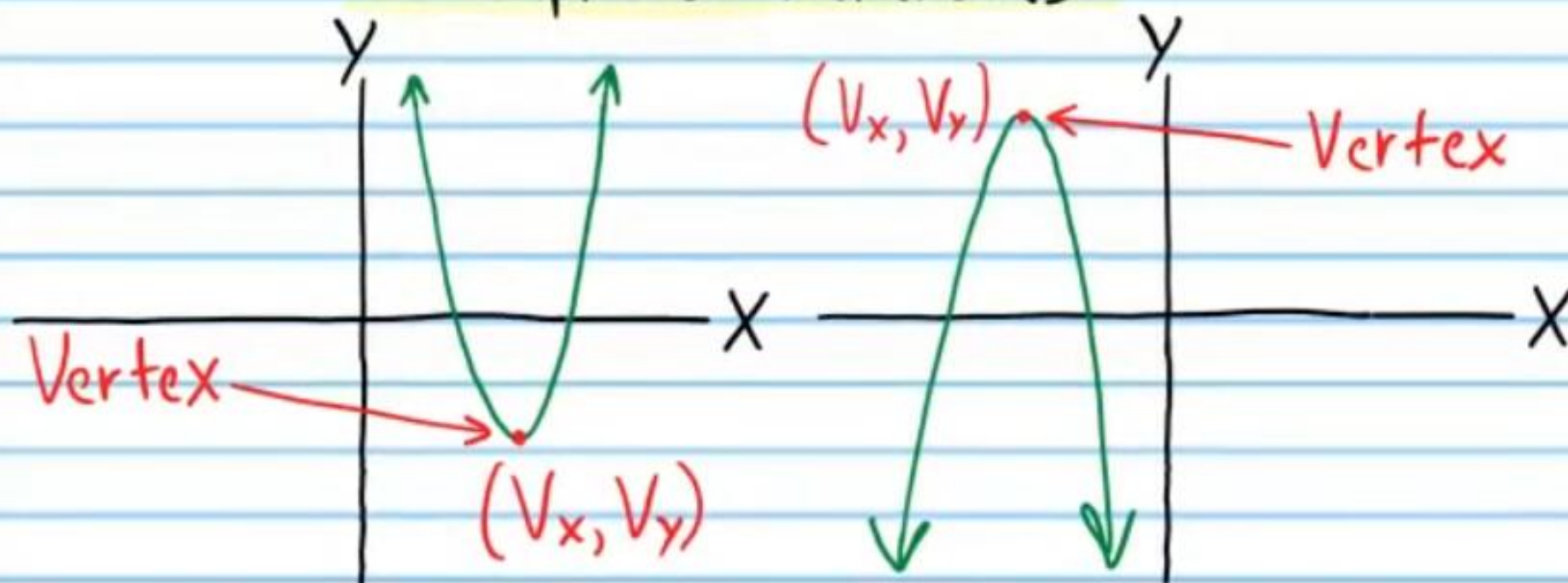
An Example Equation of a Quadratic Graph

$$0 = 6x^2 + 2x - 5$$

$$y = 6x^2 + 2x - 5$$

Graphs of Quadratic Expressions (called parabolas)

Examples of Parabolas

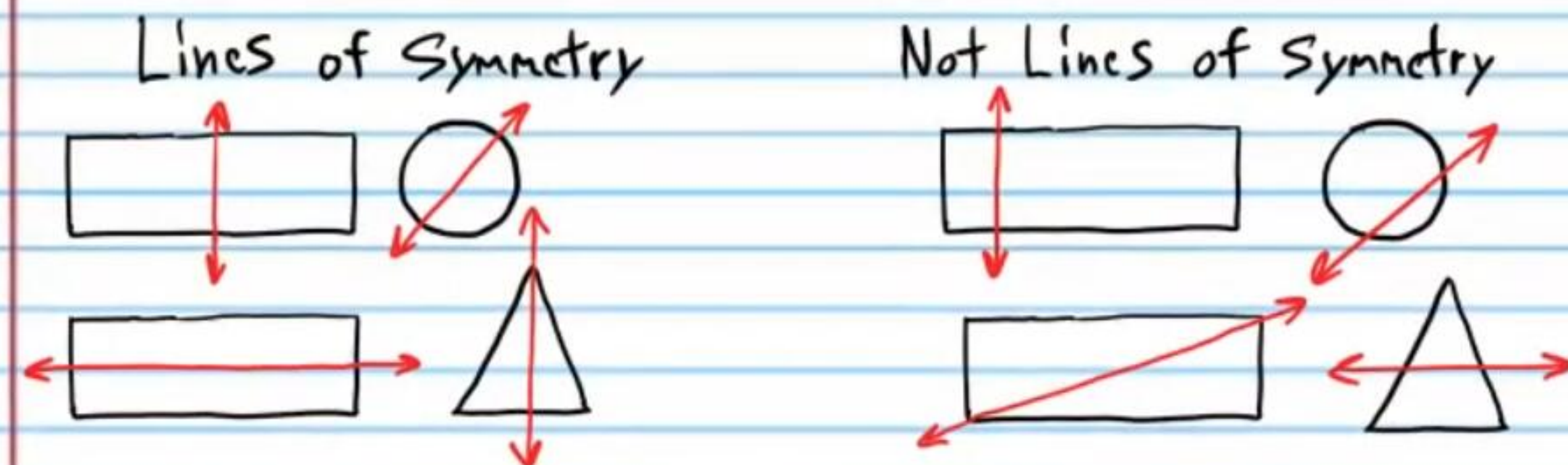


The Three Forms of a Quadratic Expression

Standard Form	Intercept Form	Vertex Form
$y = ax^2 + bx + c$	$y = a(x - m)(x - n)$	$y = a(x - h)^2 + k$

Preparation

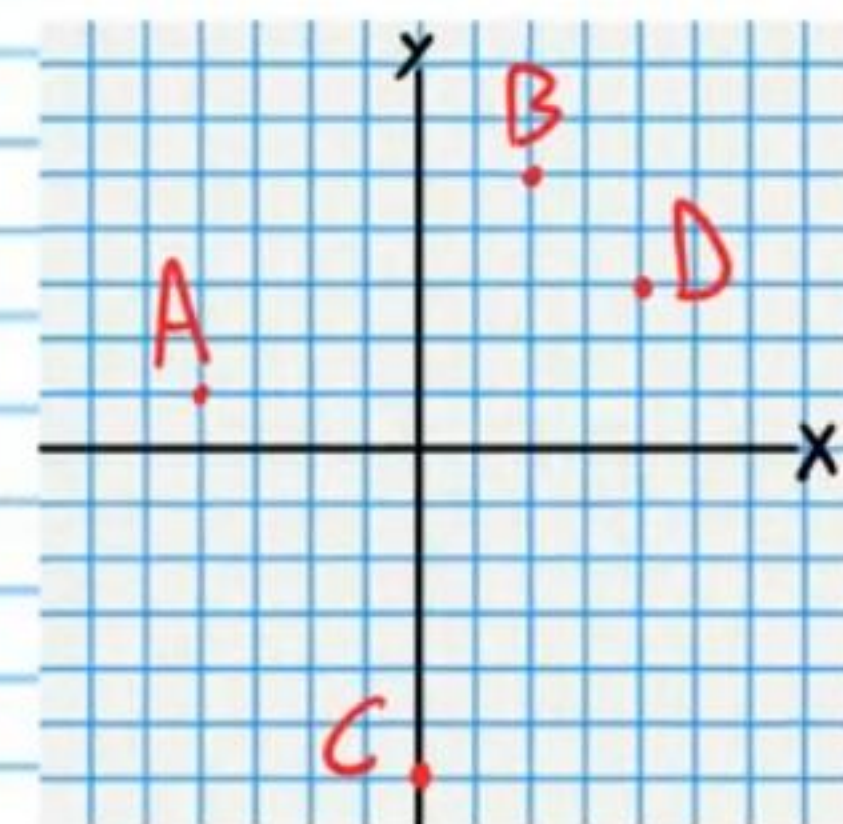
Symmetry



Plotting Ordered Pairs with Tables

Plot the points A(-4, 1), B(2, 5), C(0, -6), and D(4, 3).

x	y
-4	1
2	5
0	-6
4	3



Evaluating Algebraic Expressions

Evaluate.

$$(x + 3)^2 - 4x + 8x^2 + \frac{x}{7} \text{ if } x = -2$$

PEMDAS

$$(-2 + 3)^2 - 4(-2) + 8(-2)^2 + \left(-\frac{2}{7}\right)$$

$$1^2 + 8 + 8(4) - \frac{2}{7}$$

$$1 + 8 + 32 - \frac{2}{7}$$

$$40 \frac{5}{7}$$

Graphing Quadratic Expressions in Standard Form

Finding the Vertex in Standard Form

If a quadratic expression is written in the form $y = ax^2 + bx + c$, then the x-value of the vertex is $-\frac{b}{2a}$.

Graph the following quadratic expressions.

① $y = 3x^2 - 12x + 15$

$a = 3, b = -12, c = 15$

I) $V_x = -\frac{b}{2a}$

$V_x = -\frac{(-12)}{2(3)}$

$V_x = \frac{12}{6}$

$V_x = 2$

$V_y = 3(2)^2 - 12(2) + 15$

$V_y = 3(4) - 12(2) + 15$

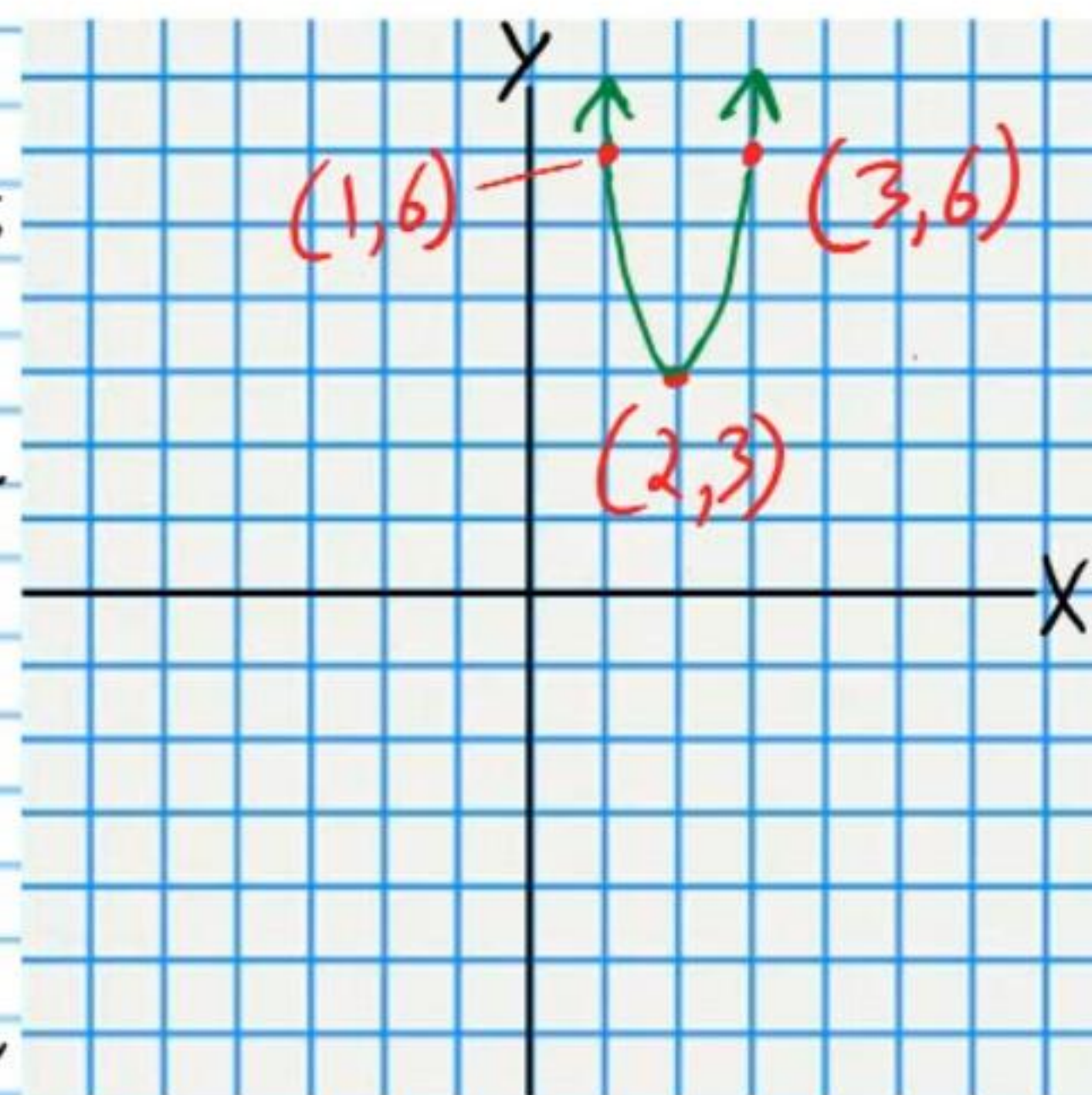
$V_y = 12 - 24 + 15 = 3$

II) $x = 3$

$y = 3(3)^2 - 12(3) + 15$

$y = 3(9) - 36 + 15$

$y = 6$



② $x^2 + 8x + 15 = y$

$a = 1, b = 8, c = 15$

I) $V_x = -\frac{b}{2a}$

$V_x = -\frac{8}{2(1)}$

$V_x = -\frac{8}{2}$

$V_x = -4$

$V_y = (-4)^2 + 8(-4) + 15$

$V_y = 16 - 32 + 15$

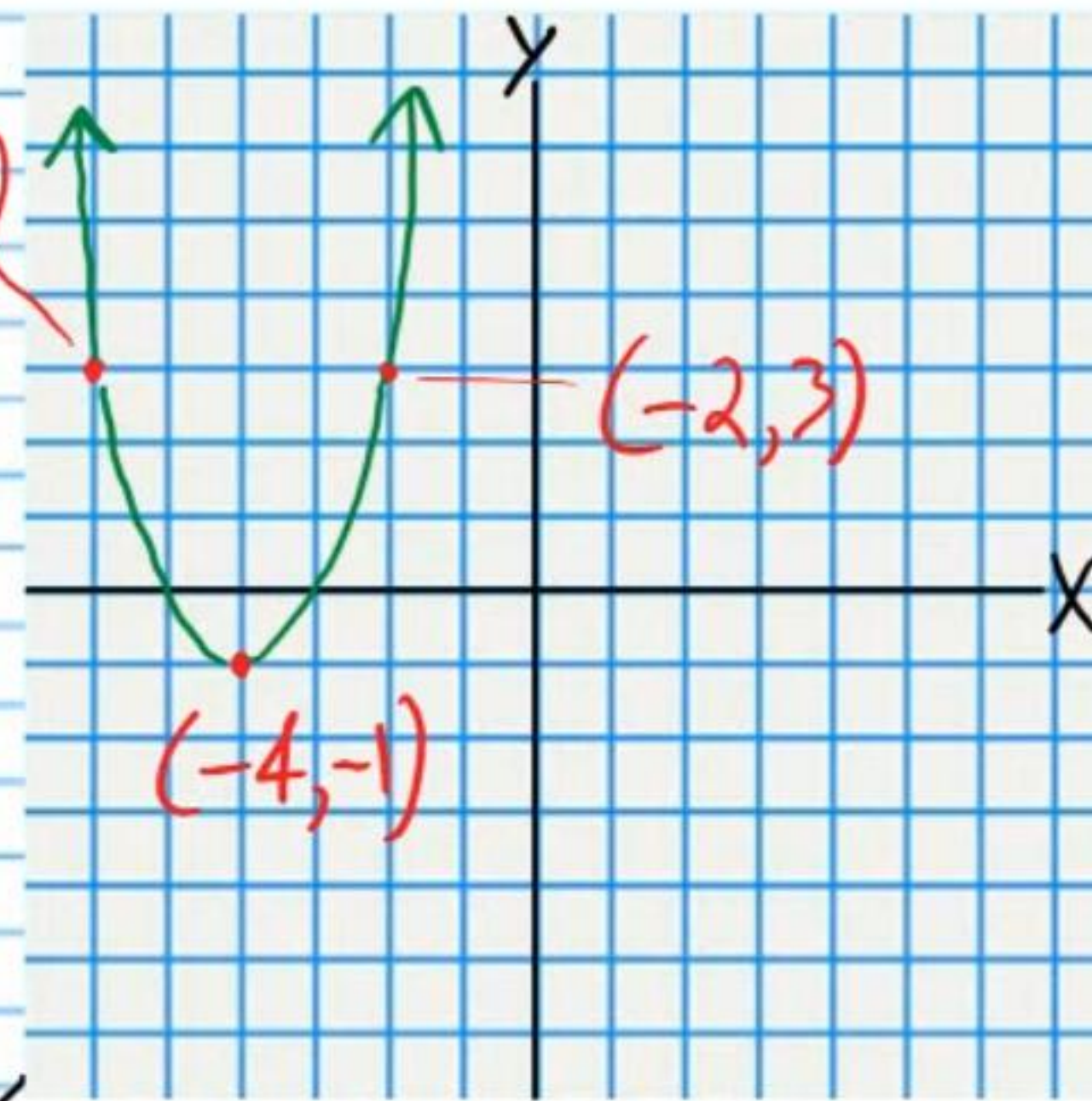
$V_y = -1$

II) $x = -2$

$y = (-2)^2 + 8(-2) + 15$

$y = 4 - 16 + 15$

$y = 3$



$$\textcircled{3} y = -2x^2 - 4x + 1$$

$$a = -2, b = -4, c = 1$$

$$\text{I) } V_x = -\frac{b}{2a}$$

$$V_x = -\frac{(-4)}{2(-2)}$$

$$V_x = \frac{4}{-4}$$

$$V_x = -1$$

$$\rightarrow V_y = -2(-1)^2 - 4(-1) + 1$$

$$V_y = -2(1) - 4(-1) + 1$$

$$V_y = -2 + 4 + 1$$

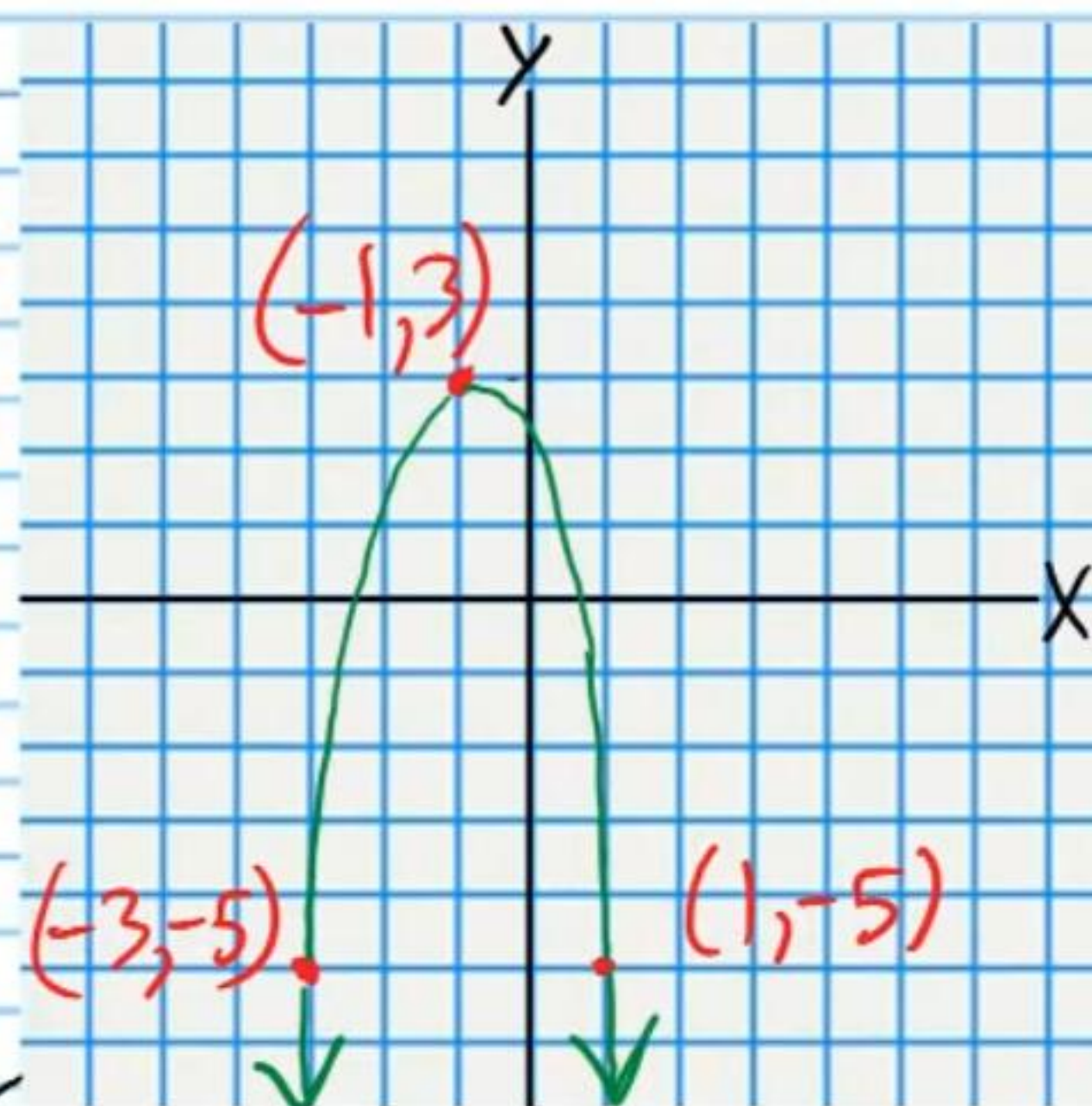
$$V_y = 3$$

$$\text{II) } x = 1$$

$$\rightarrow y = -2(1)^2 - 4(1) + 1$$

$$y = -2(1) - 4(1) + 1$$

$$y = -2 - 4 + 1 = -5$$



$$\textcircled{4} 5x^2 + 30x + 43 = y$$

$$a = 5, b = 30, c = 43$$

$$\text{I) } V_x = -\frac{b}{2a}$$

$$V_x = -\frac{30}{2(5)}$$

$$V_x = -\frac{30}{10}$$

$$V_x = -3$$

$$\rightarrow V_y = 5(-3)^2 + 30(-3) + 43$$

$$V_y = 5(9) + 30(-3) + 43$$

$$V_y = 45 - 90 + 43$$

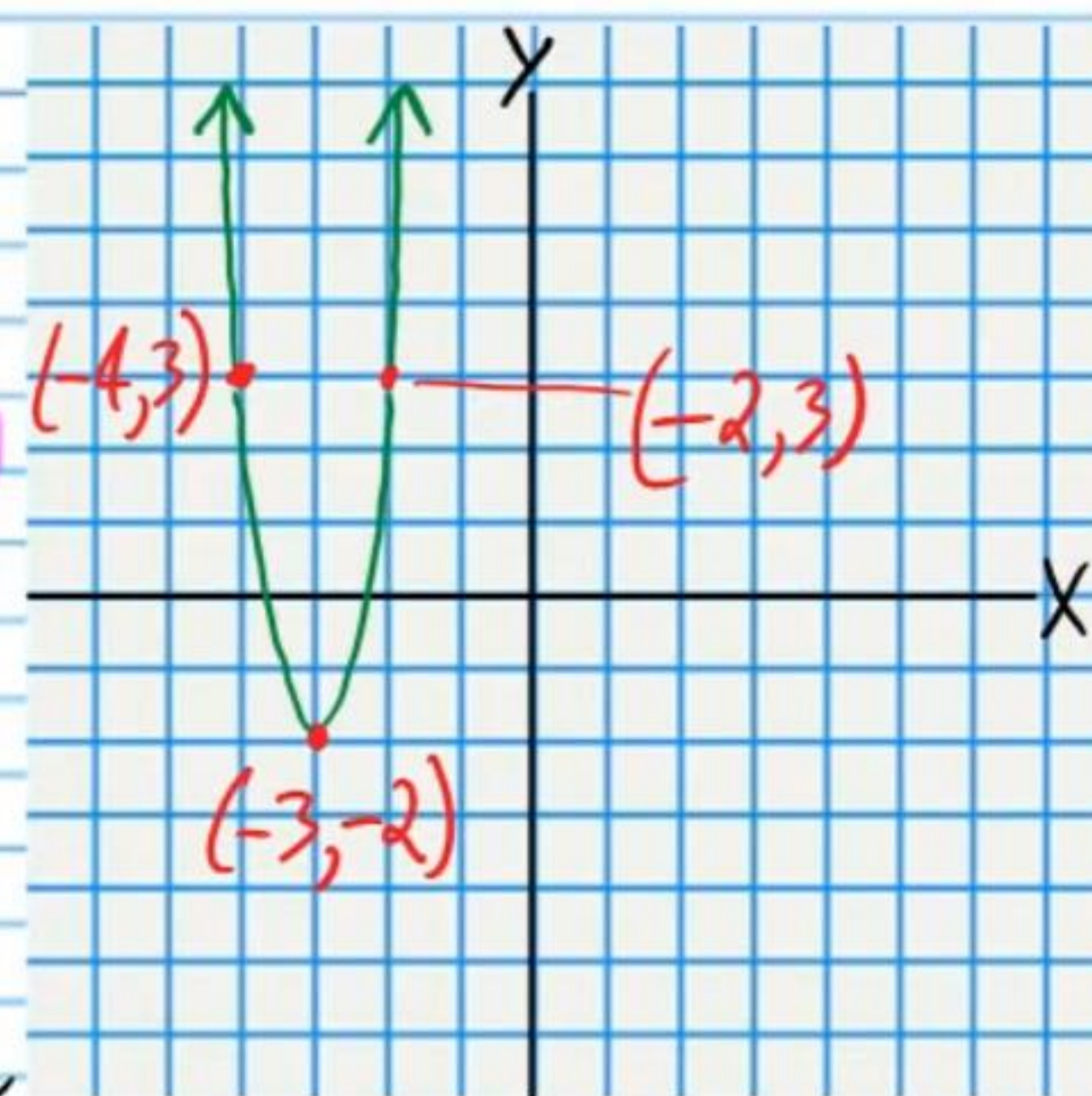
$$V_y = -2$$

$$\text{II) } x = -2$$

$$\rightarrow y = 5(-2)^2 + 30(-2) + 43$$

$$y = 5(4) + 30(-2) + 43$$

$$y = 20 - 60 + 43 = 3$$



$$⑤ y = x^2 - 8x + 11$$

$$a=1, b=-8, c=11$$

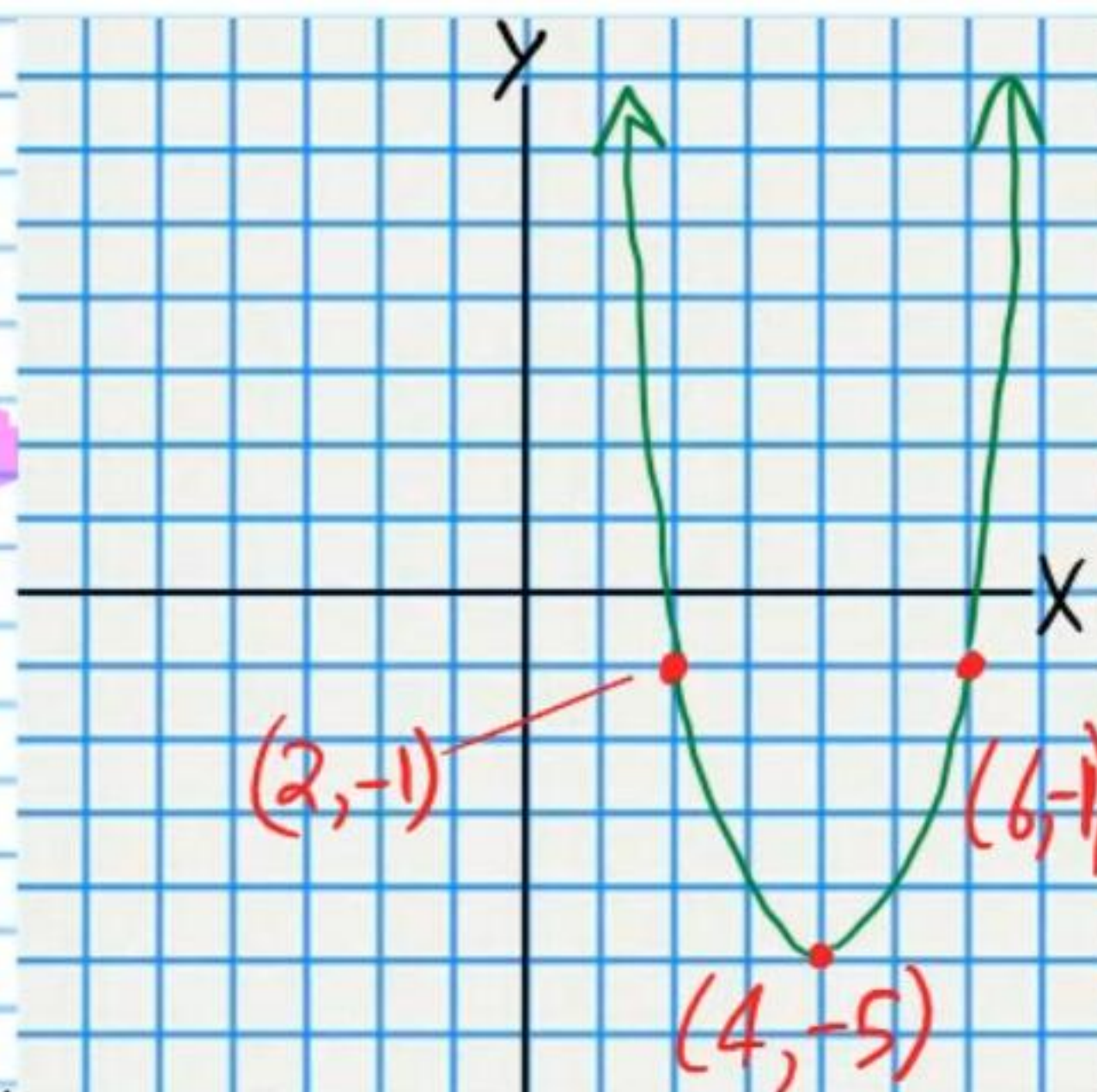
$$I) V_x = -\frac{b}{2a}$$

x	y
4	-5
2	-1
6	-1

$$V_x = -\frac{(-8)}{2(1)}$$

$$V_x = \frac{8}{2}$$

$$V_x = 4$$



$$V_y = 4^2 - 8(4) + 11$$

$$V_y = 16 - 32 + 11$$

$$V_y = -5$$

$$II) x=2$$

$$y = (2)^2 - 8(2) + 11$$

$$y = 4 - 16 + 11$$

$$y = -1$$

$$⑥ -3x^2 - 12x - 12 = y$$

$$a=-3, b=-12, c=-12$$

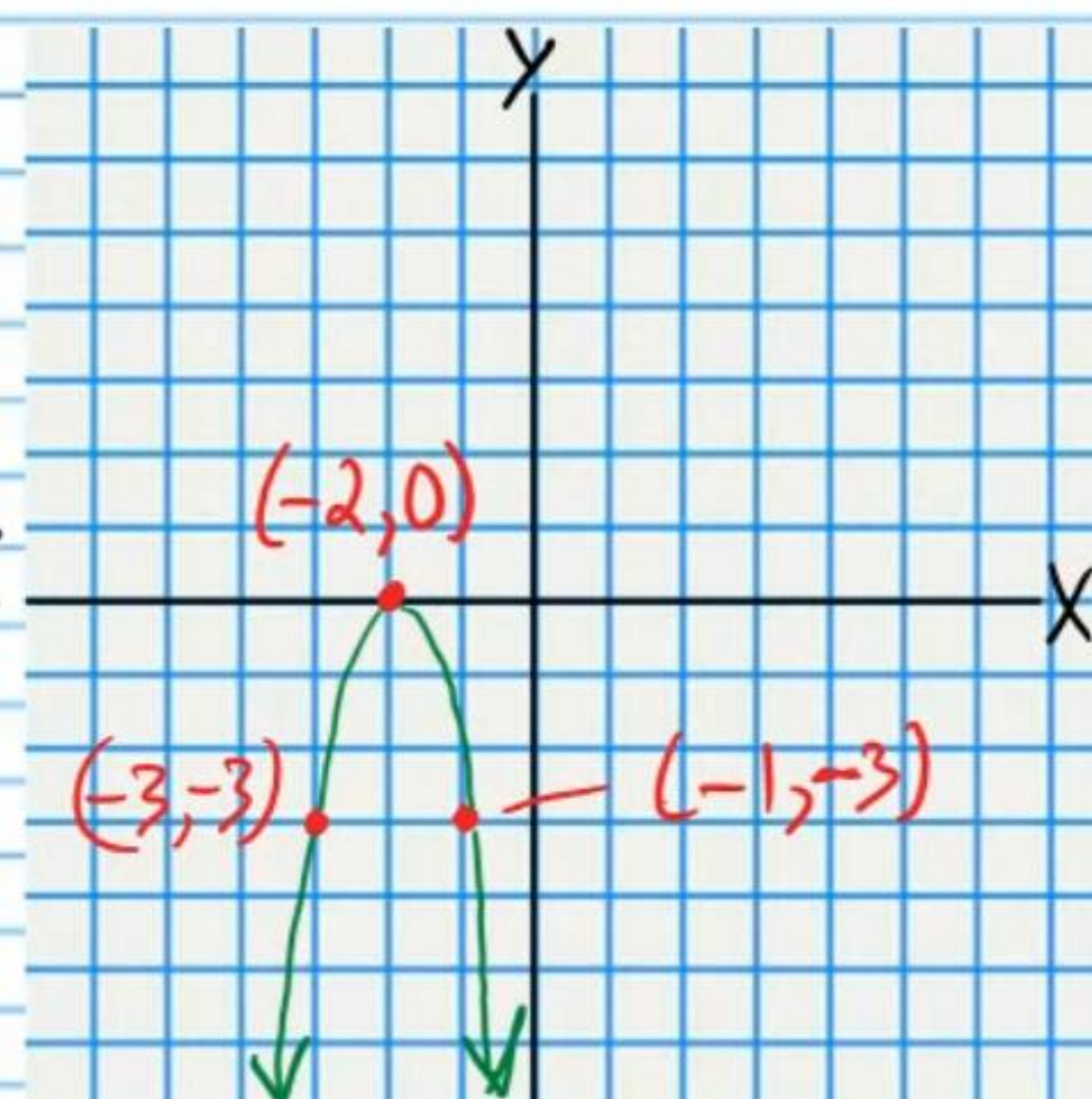
$$I) V_x = -\frac{b}{2a}$$

x	y
-2	0
-1	-3
-3	-3

$$V_x = -\frac{(-12)}{2(-3)}$$

$$V_x = \frac{12}{-6}$$

$$V_x = -2$$



$$V_y = -3(-2)^2 - 12(-2) - 12$$

$$V_y = -3(4) - 12(-2) - 12$$

$$V_y = -12 + 24 - 12$$

$$V_y = 0$$

$$II) x=-1$$

$$y = -3(-1)^2 - 12(-1) - 12$$

$$y = -3(1) + 12 - 12 = -3$$

X and y Intercepts

Find the x and y intercepts of each equation below and use them to draw the corresponding graphs.

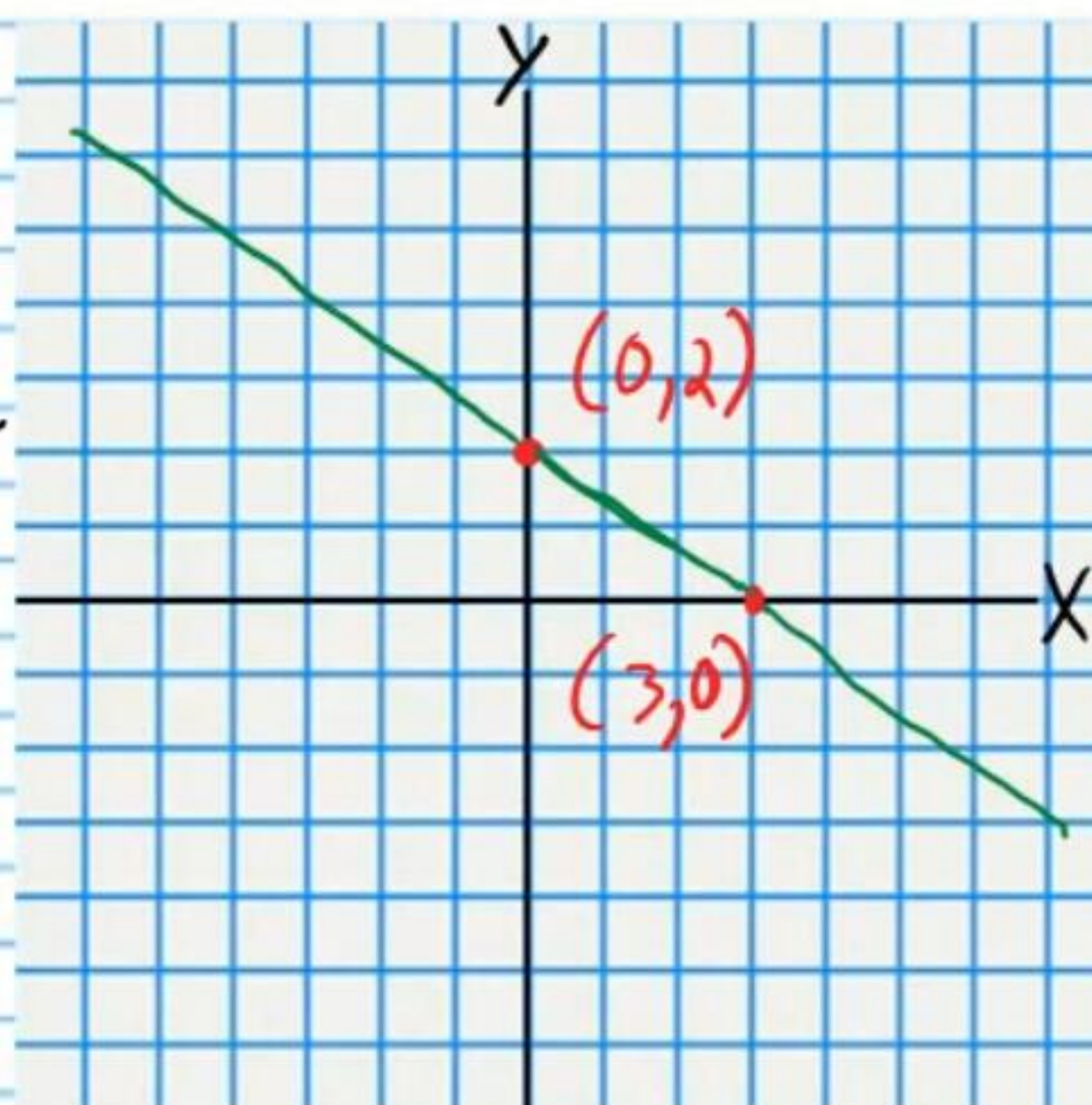
Example I)

$$2x + 3y = 6$$

x	y
3	0
0	2

I) X-Intercept
 $y = 0$
 $2x + 3(0) = 6$
 $x = 3$

II) Y-Intercept
 $x = 0$
 $2(0) + 3y = 6$
 $y = 2$



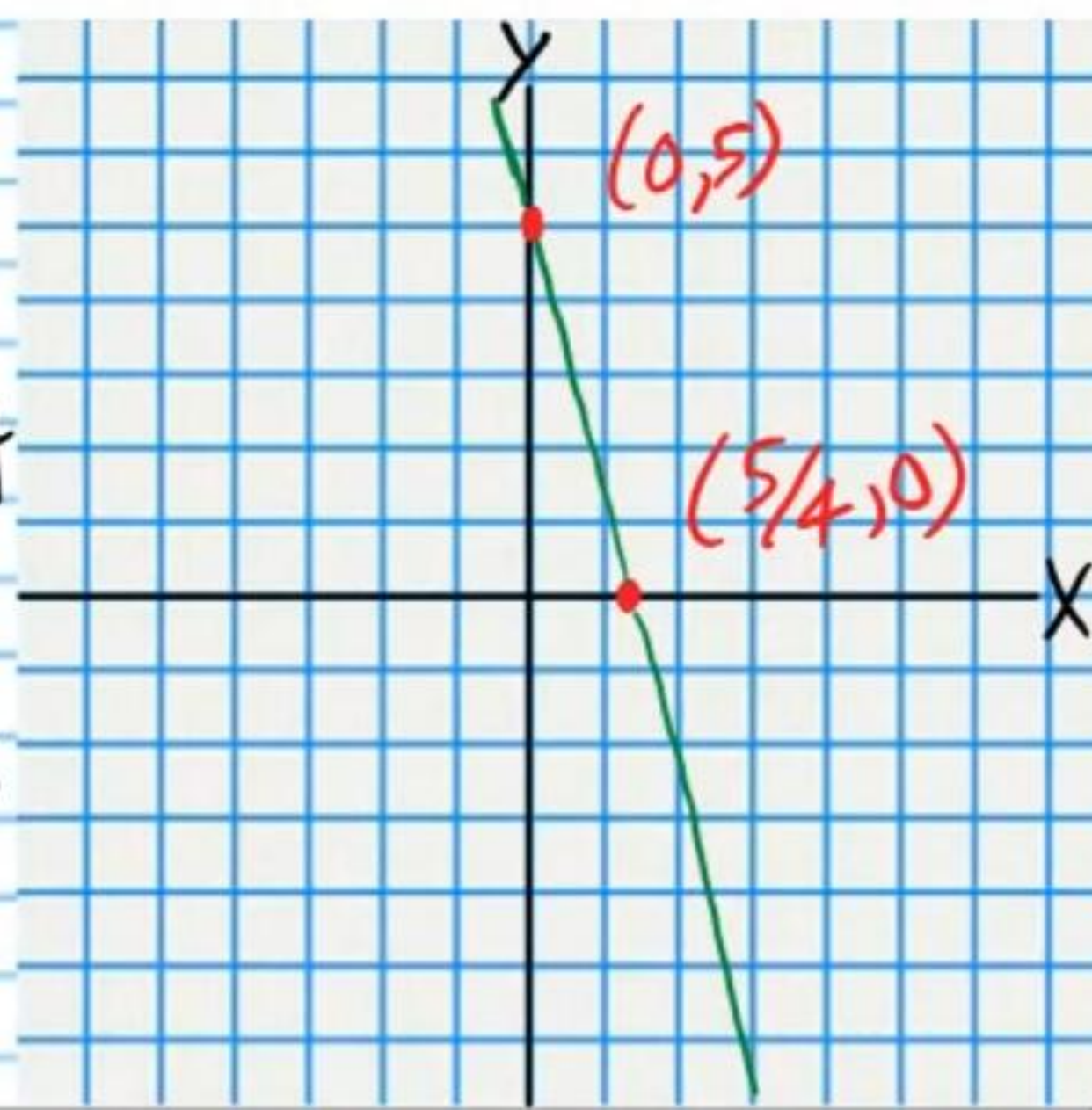
Example II)

$$y = -4x + 5$$

x	y
0	5
5/4	0

I) X-Intercept
 $x = 0$
 $y = -4(0) + 5$
 $y = 5$

II) Y-Intercept
 $y = 0$
 $0 = -4x + 5$
 $\frac{5}{4} = x$



Graphing Quadratic Expressions in Intercept Form

Finding the Vertex in Intercept Form

If a quadratic expression is written in the form $y = a(x-m)(x-n)$, where m and n are real numbers, then the x-value of the vertex is $\frac{m+n}{2}$.

* Intercept form cannot be used to graph a quadratic expression if the corresponding graph does not move through the x-axis.

Graph the following quadratic expressions.

⑦ $y = 3(x-2)(x-6)$

$a = 3, m = 2, n = 6$

I) $(2, 0)$ & $(6, 0)$

x	y
4	-12
2	0
6	0

II) $V_x = \frac{m+n}{2}$ $V_y = 3(4-2)(4-6)$

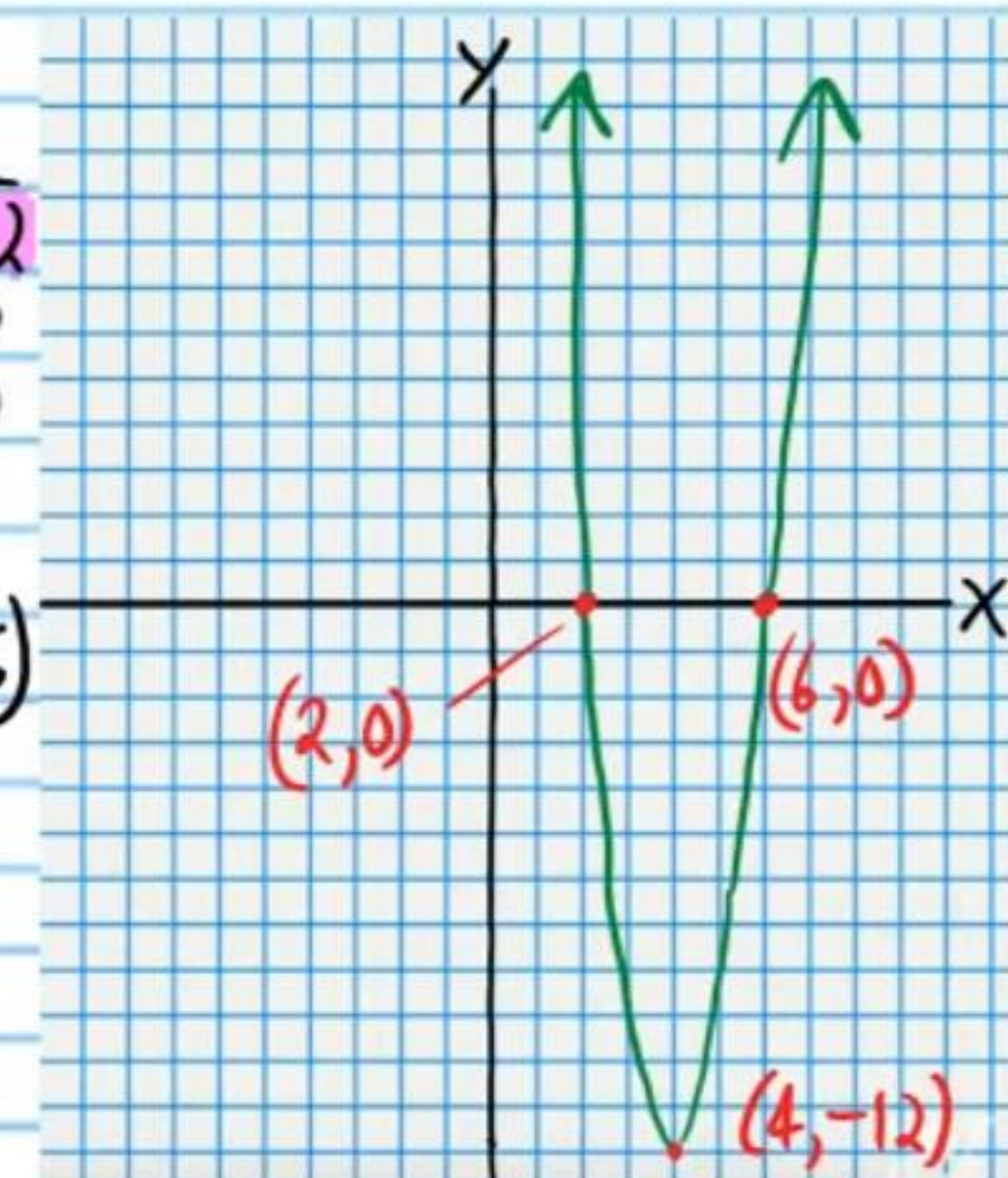
$V_x = \frac{2+6}{2}$

$V_x = \frac{8}{2}$

$V_x = 4$

$V_y = 3(2)(-2)$

$V_y = -12$



⑧ $-2(x+4)(x-2)=y$

$a=-2, m=-4, n=2$

I) $(-4,0)$ & $(2,0)$

x	y
-1	18
-4	0
2	0

II) $V_x = \frac{m+n}{2}$

$V_x = \frac{-4+2}{2}$

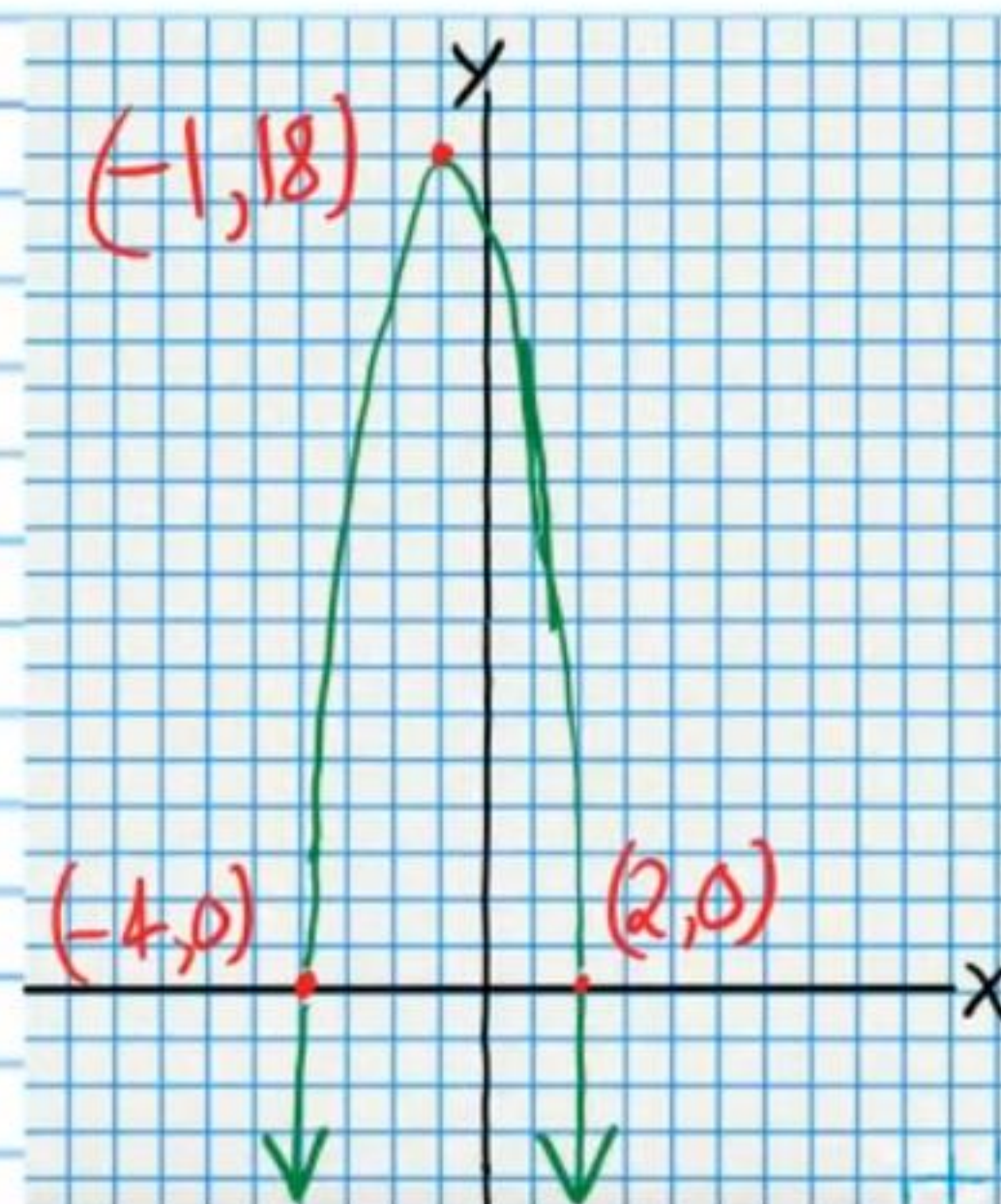
$V_x = \frac{-2}{2}$

$V_x = -1$

→ $V_y = -2(-1+4)(-1-2)$

$V_y = -2(3)(-3)$

$V_y = 18$



⑨ $y = (x-1)(x+4)$

$a=1, m=1, n=-4$

I) $(1,0)$ & $(-4,0)$

x	y
-1.5	-6.25
1	0
-4	0

II) $V_x = \frac{m+n}{2}$

$V_x = \frac{1+(-4)}{2}$

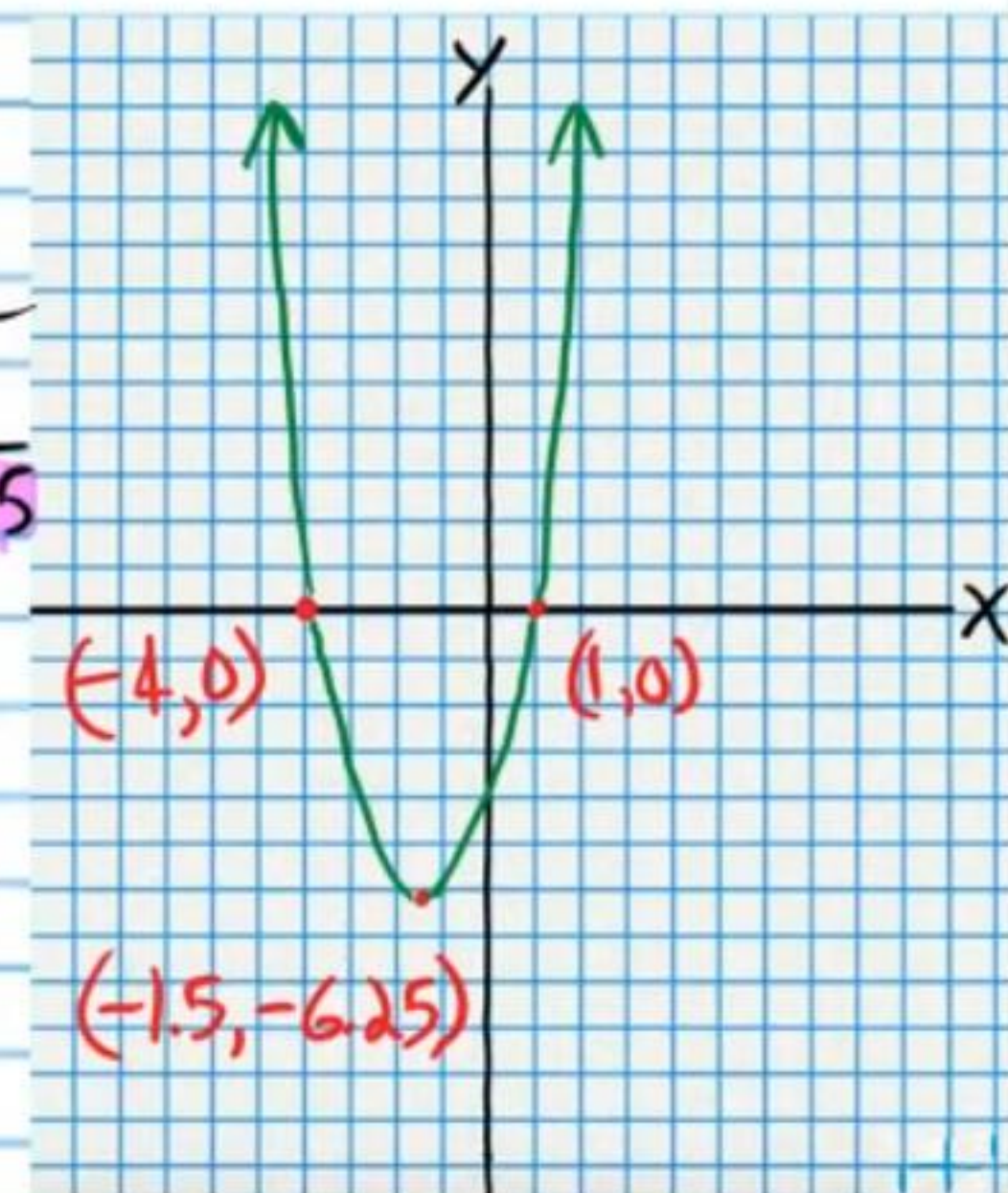
$V_x = -\frac{3}{2}$

$V_x = -1.5$

→ $V_y = (-1.5-1)(-1.5+4)$

$V_y = (-2.5)(2.5)$

$V_y = -6.25$



$$\textcircled{10} -4(x+1)(x-3)=y$$

$$a=-4, m=-1, n=3$$

$$\text{I) } (-1,0) \text{ \& } (3,0)$$

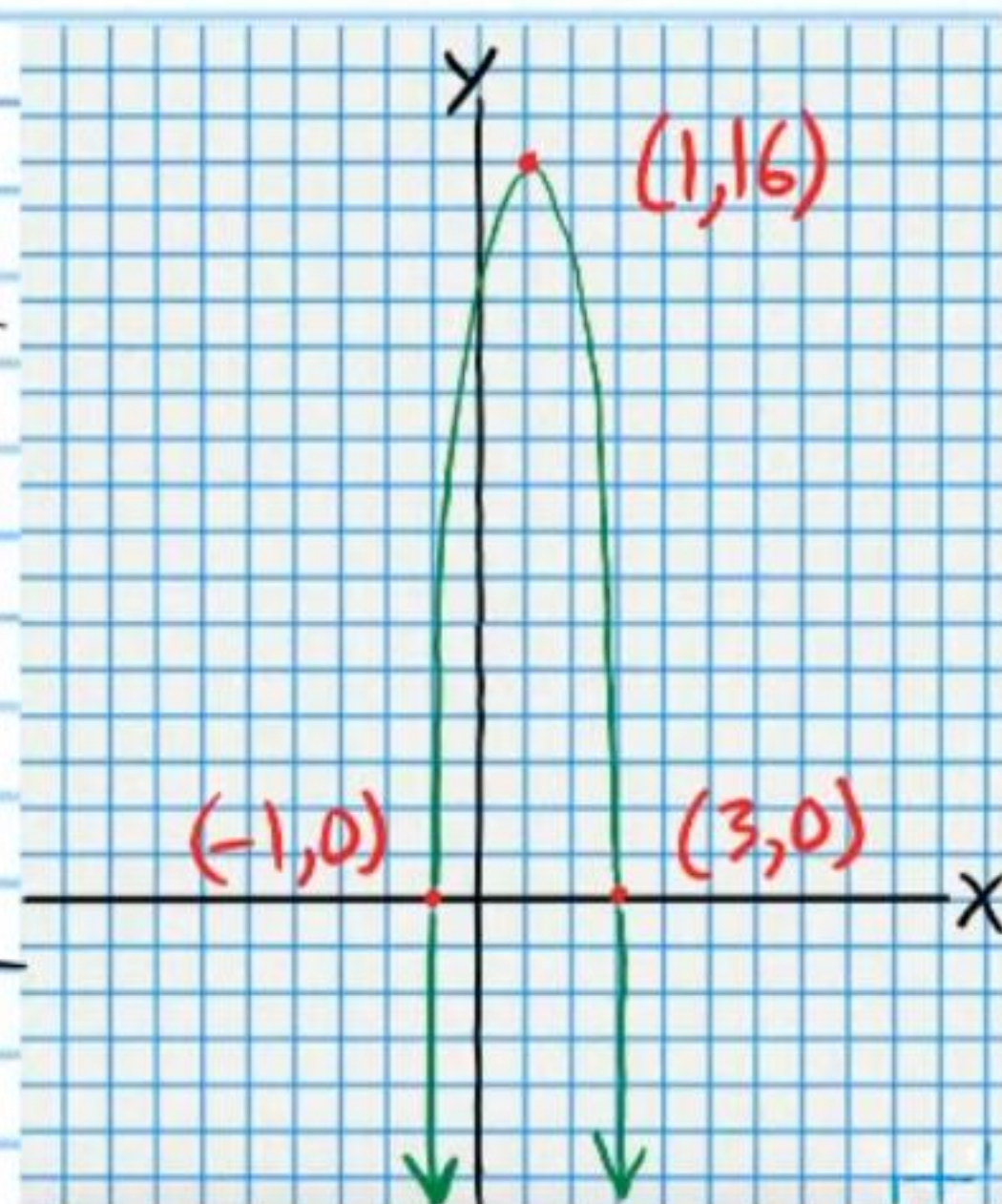
$$\text{II) } v_x = \frac{m+n}{2}$$

$$v_x = \frac{-1+3}{2}$$

$$v_x = \frac{2}{2}$$

$$v_x = 1$$

x	y
1	16
-1	0
3	0



$$\rightarrow v_y = -4(1+1)(1-3)$$

$$v_y = -4(2)(-2)$$

$$v_y = 16$$

$$\textcircled{11} y = -(x+1)(x+9)$$

$$a=-1, m=-1, n=-9$$

$$\text{I) } (-1,0) \text{ \& } (-9,0)$$

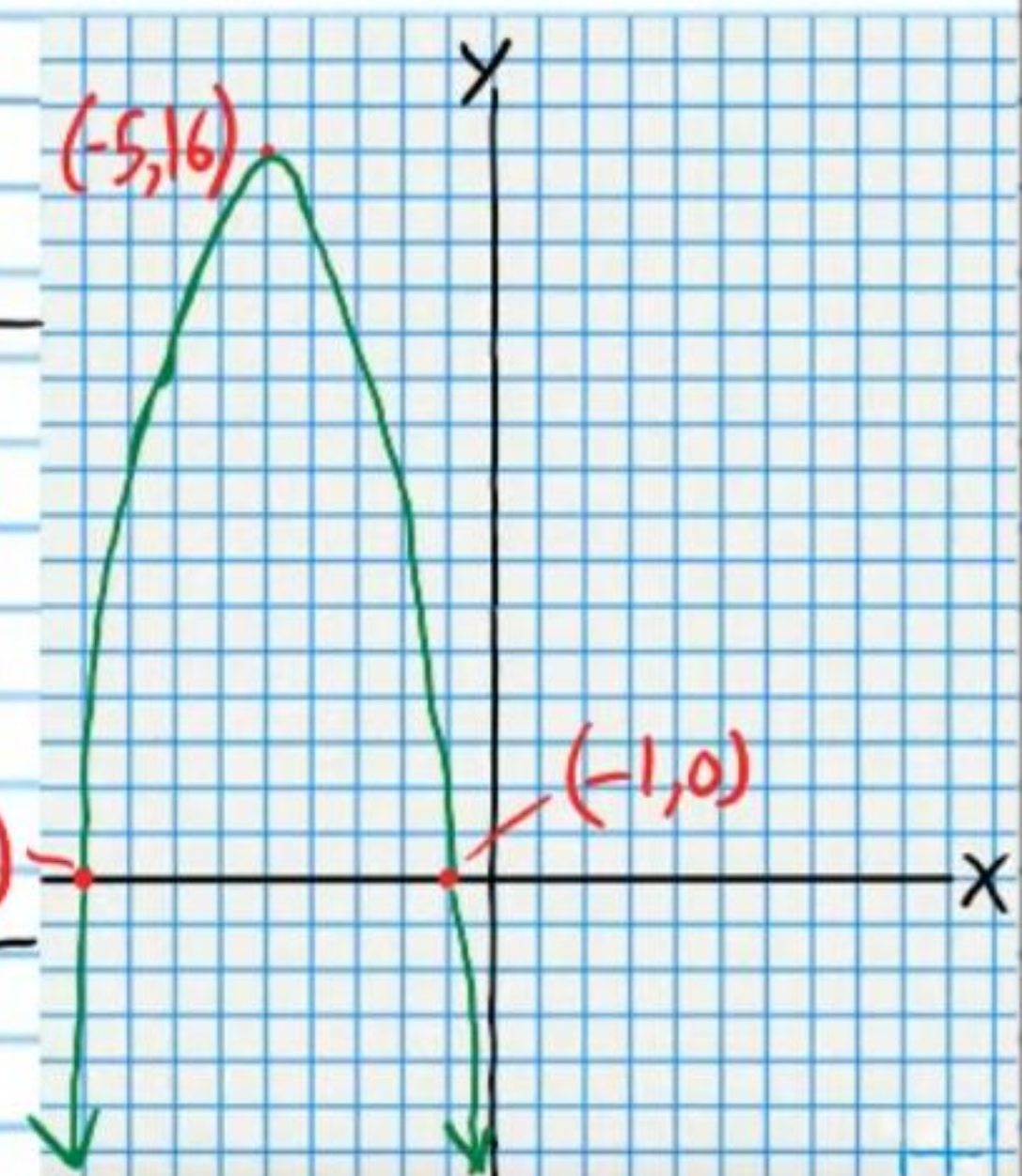
$$\text{II) } v_x = \frac{m+n}{2}$$

$$v_x = \frac{-1+(-9)}{2}$$

$$v_x = \frac{-10}{2}$$

$$v_x = -5$$

x	y
-5	16
-1	0
-9	0



$$\rightarrow v_y = -(-5+1)(-5+9)$$

$$v_y = -(-4)(4)$$

$$v_y = 16$$

⑫ $2(x-5)(x-6)=y$

$a=2, m=5, n=6$

I) $(5,0) \text{ \& } (6,0)$

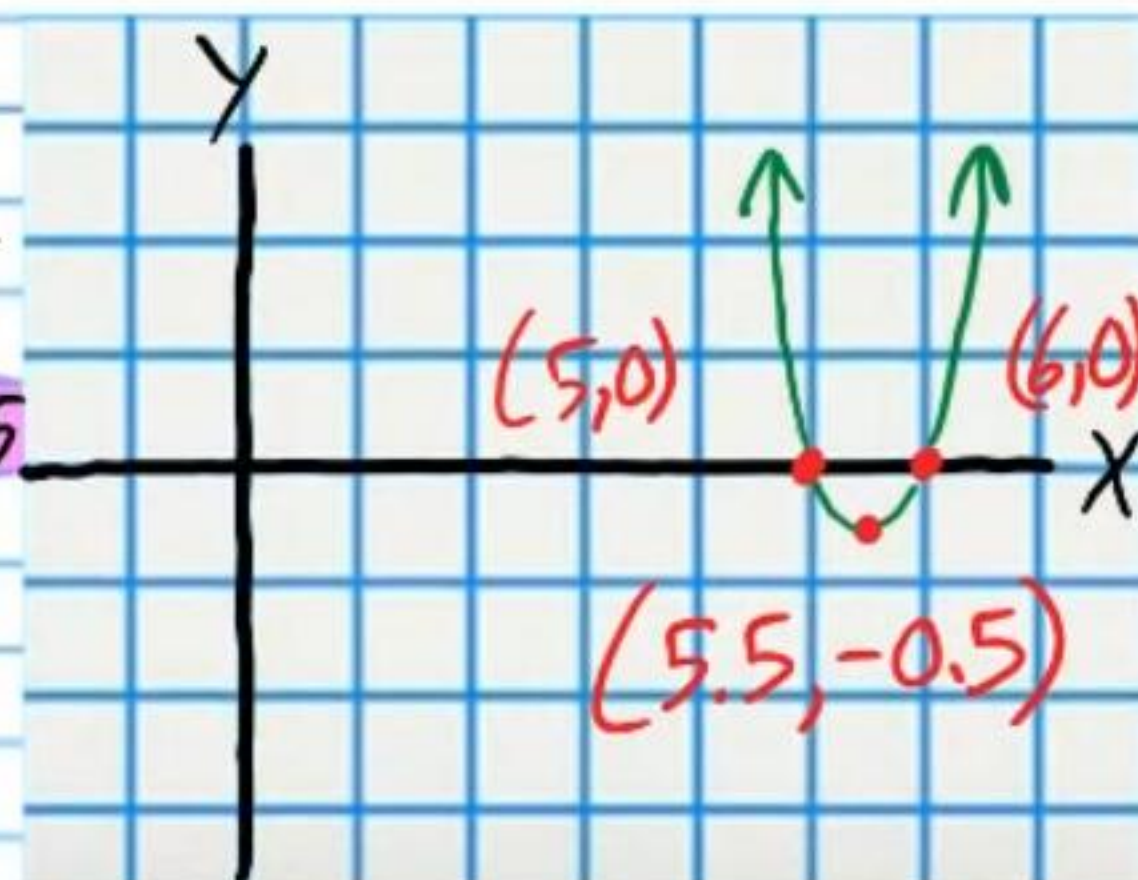
II) $V_x = \frac{m+n}{2}$

$V_x = \frac{5+6}{2}$

$V_x = \frac{11}{2}$

$V_x = 5.5$

x	y
5.5	-0.5
5	0
6	0



$V_y = 2(5.5-5)(5.5-6)$

$V_y = 2(0.5)(-0.5)$

$V_y = -\frac{1}{2}$

Graphing Quadratic Expressions in Vertex Form

Finding the Vertex in Vertex Form

If a quadratic expression is written in the form $y = a(x-h)^2 + k$, then (h, k) is the vertex.

Graph the following quadratic expressions.

⑬ $y = 2(x-3)^2 + 4$

$a=2, h=3, k=4$

I) $V_x = h = 3$

$V_y = k = 4$

$(3, 4)$

II) $x = 2$

$y = 2(2-3)^2 + 4$

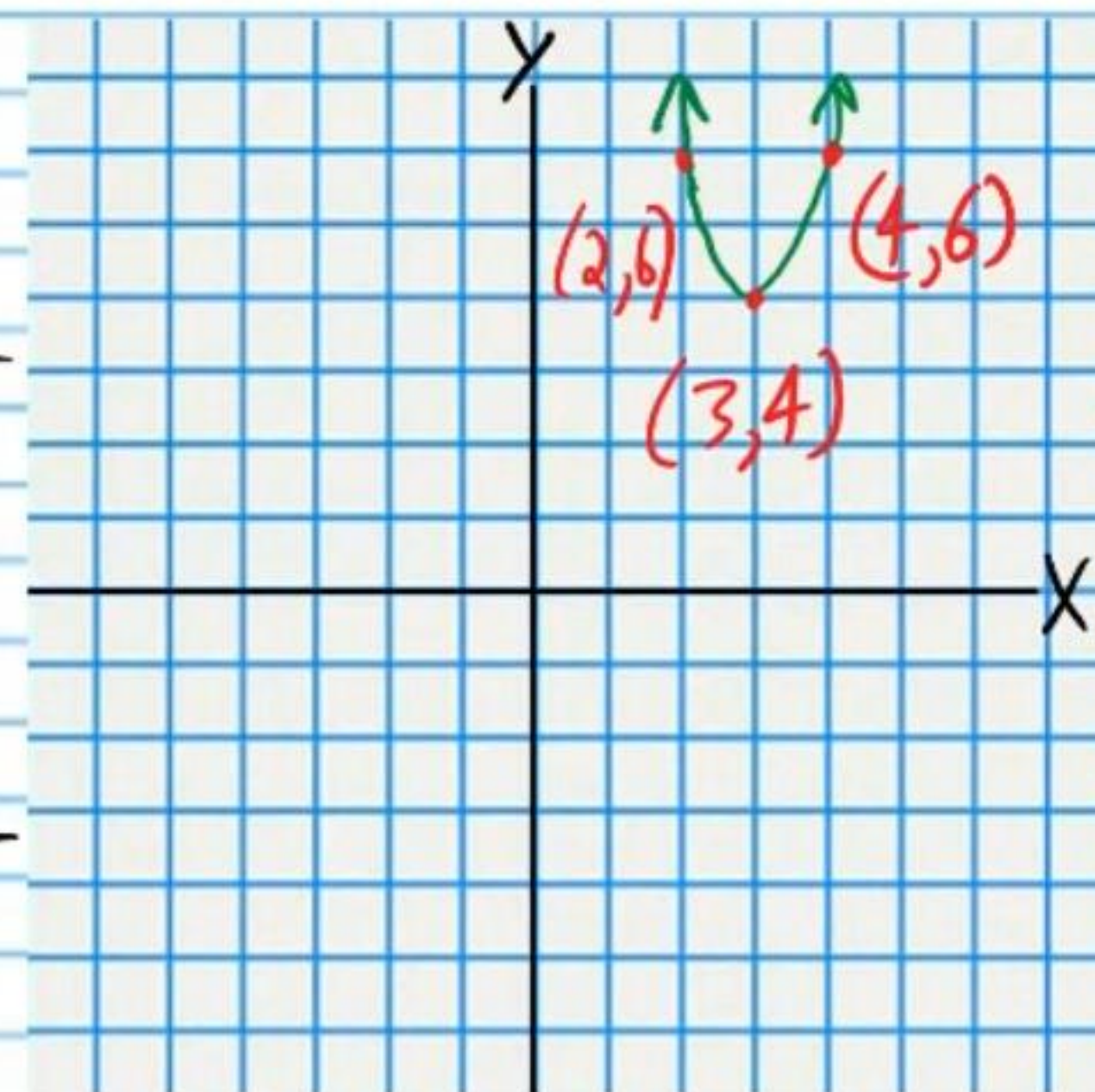
$y = 2(-1)^2 + 4$

$y = 2(1) + 4$

$y = 2 + 4$

$y = 6$

x	y
3	4
2	6
4	6



⑭ $-(x+1)^2+2=y$

$a=-1, h=-1, k=2$

I) $(-1, 2)$

X	Y
-1	2
1	-2
-3	-2

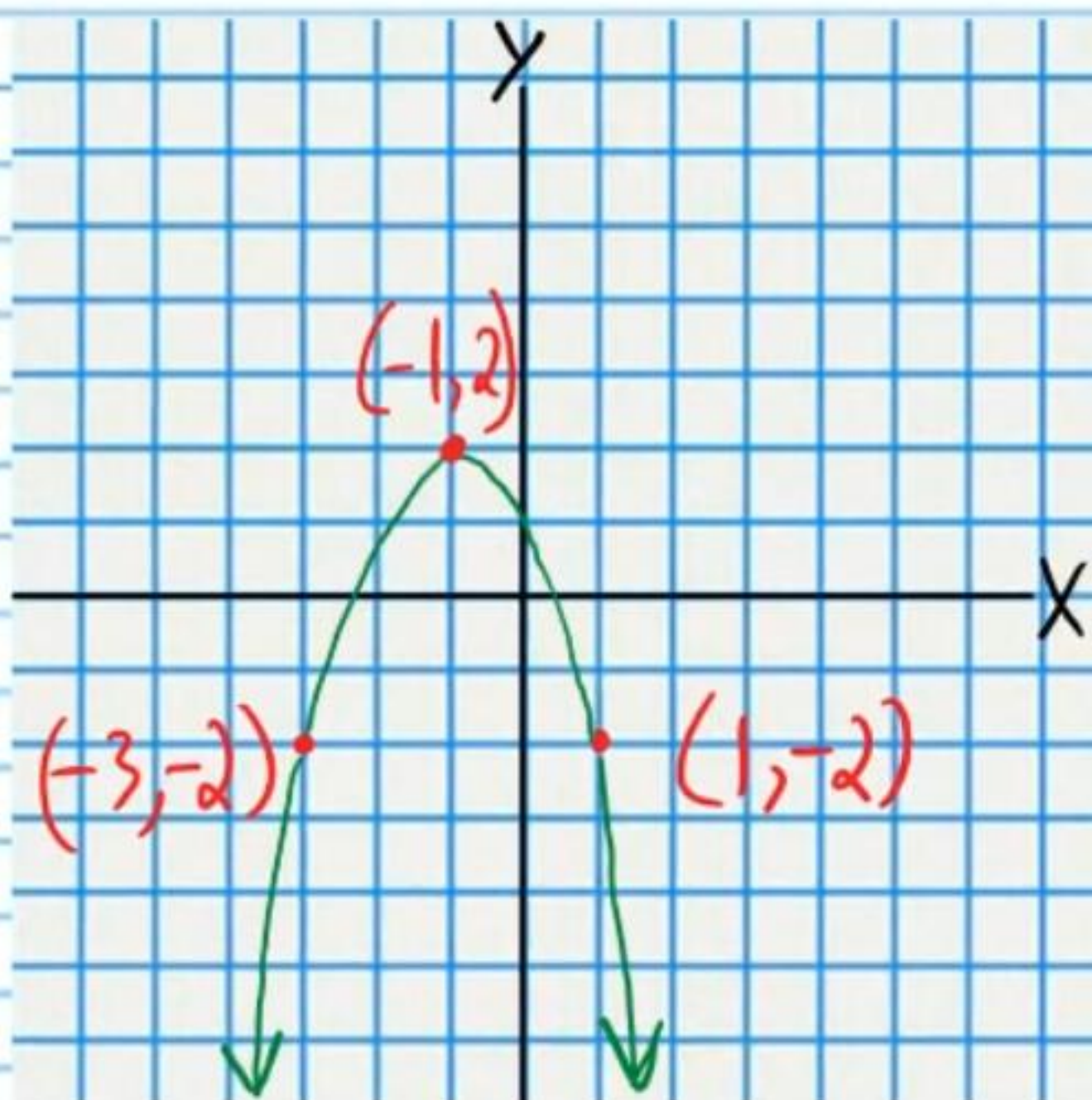
II) $x=1$

$y=-(1+1)^2+2$

$y=-(2)^2+2$

$y=-4+2$

$y=-2$



⑮ $y=5(x-4)^2-6$

$a=5, h=4, k=-6$

I) $(4, -6)$

X	Y
4	-6
3	-1
5	-1

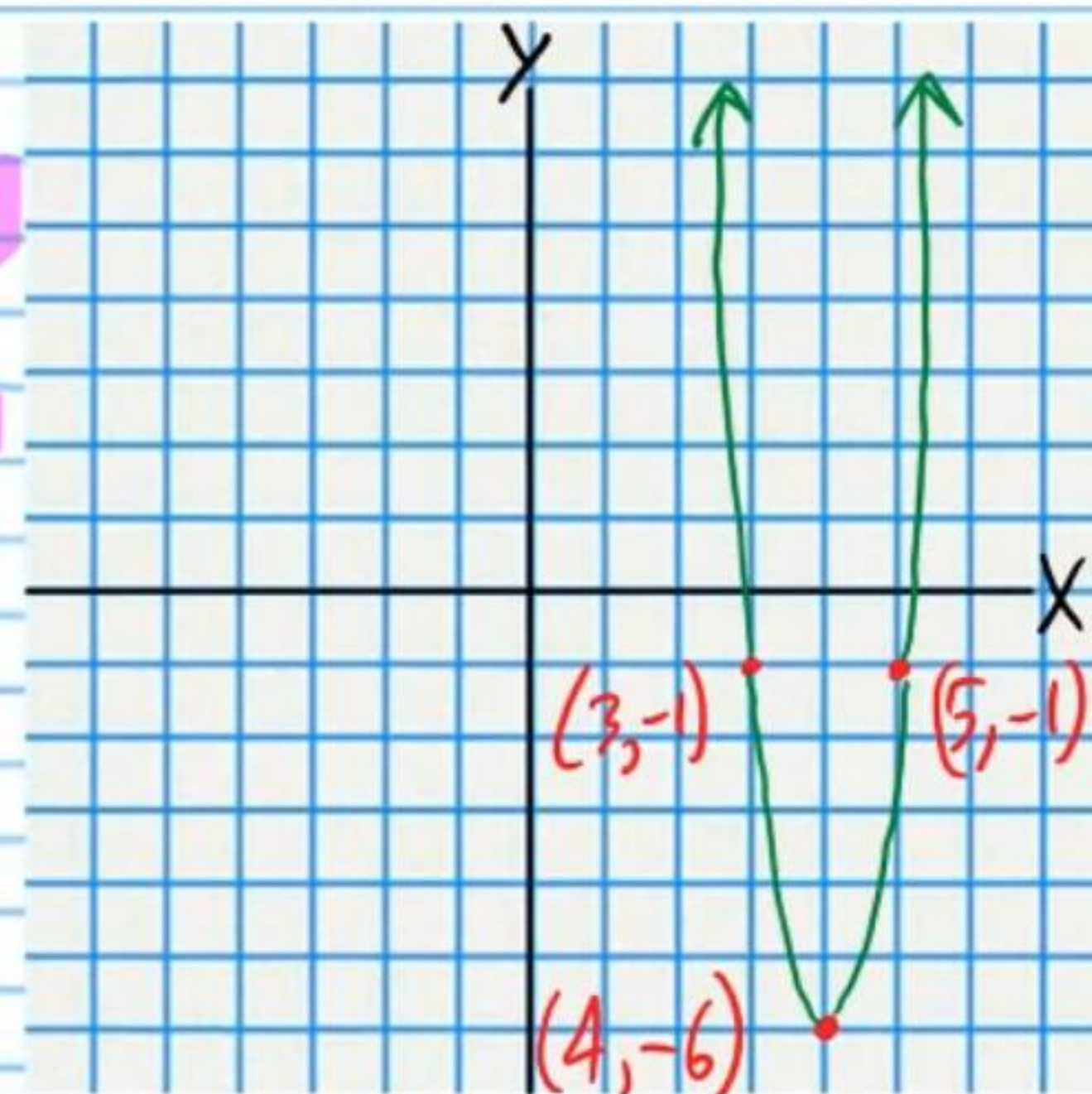
II) $x=3$

$y=5(3-4)^2-6$

$y=5(-1)^2-6$

$y=5(1)-6$

$y=5-6=-1$



⑯ $-4(x+5)^2+2=y$

$a=-4, h=-5, k=2$

I) $(-5, 2)$

X	Y
-5	2
-4	-2
-6	-2

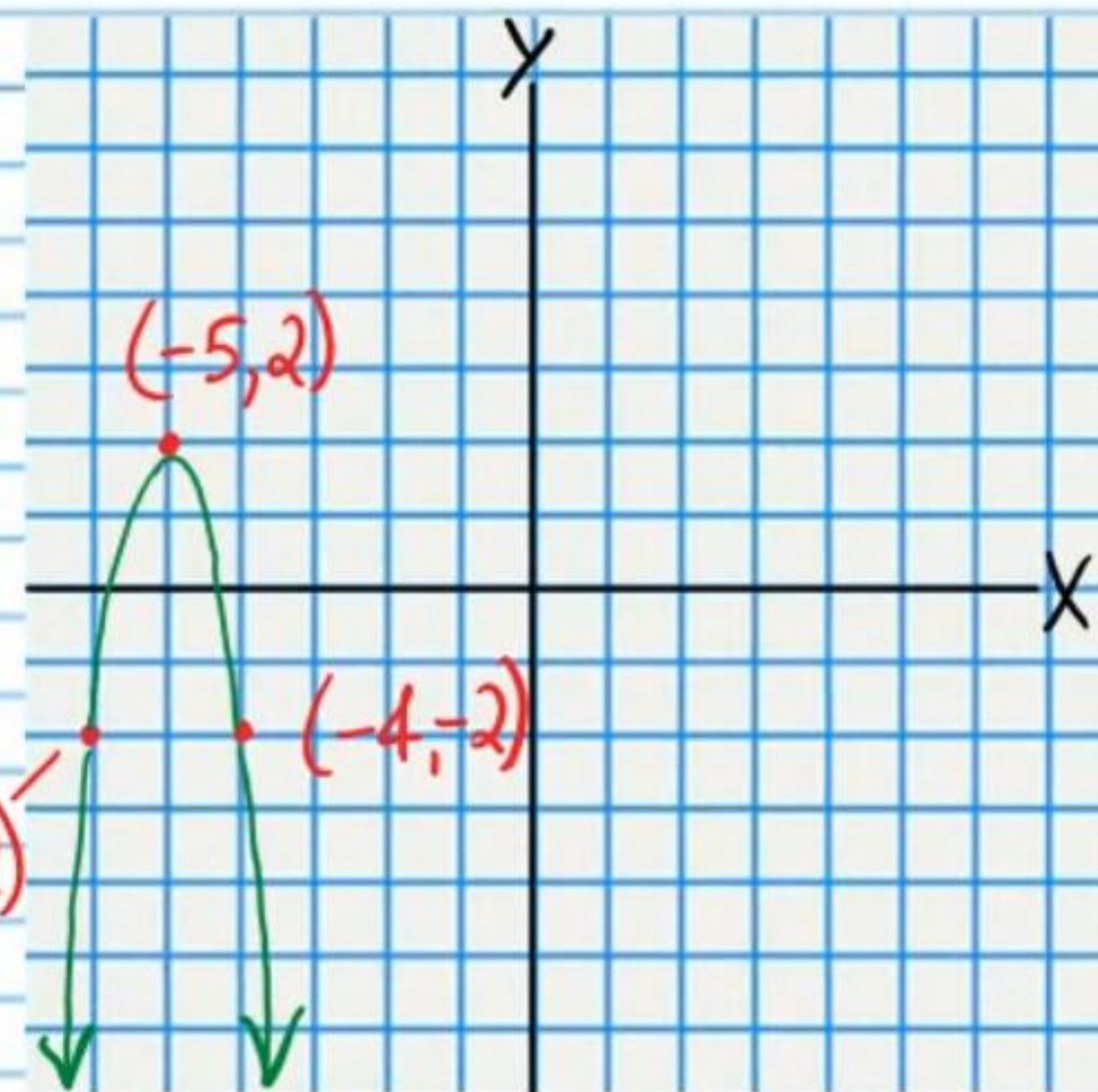
II) $x=-4$

$y=-4(-4+5)^2+2$

$y=-4(1)^2+2$

$y=-4(1)+2$

$y=-4+2=-2$



⑰ $y=-2(x-2)^2-1$

$a=-2, h=2, k=-1$

I) $(2, -1)$

X	Y
2	-1
1	-3
3	-3

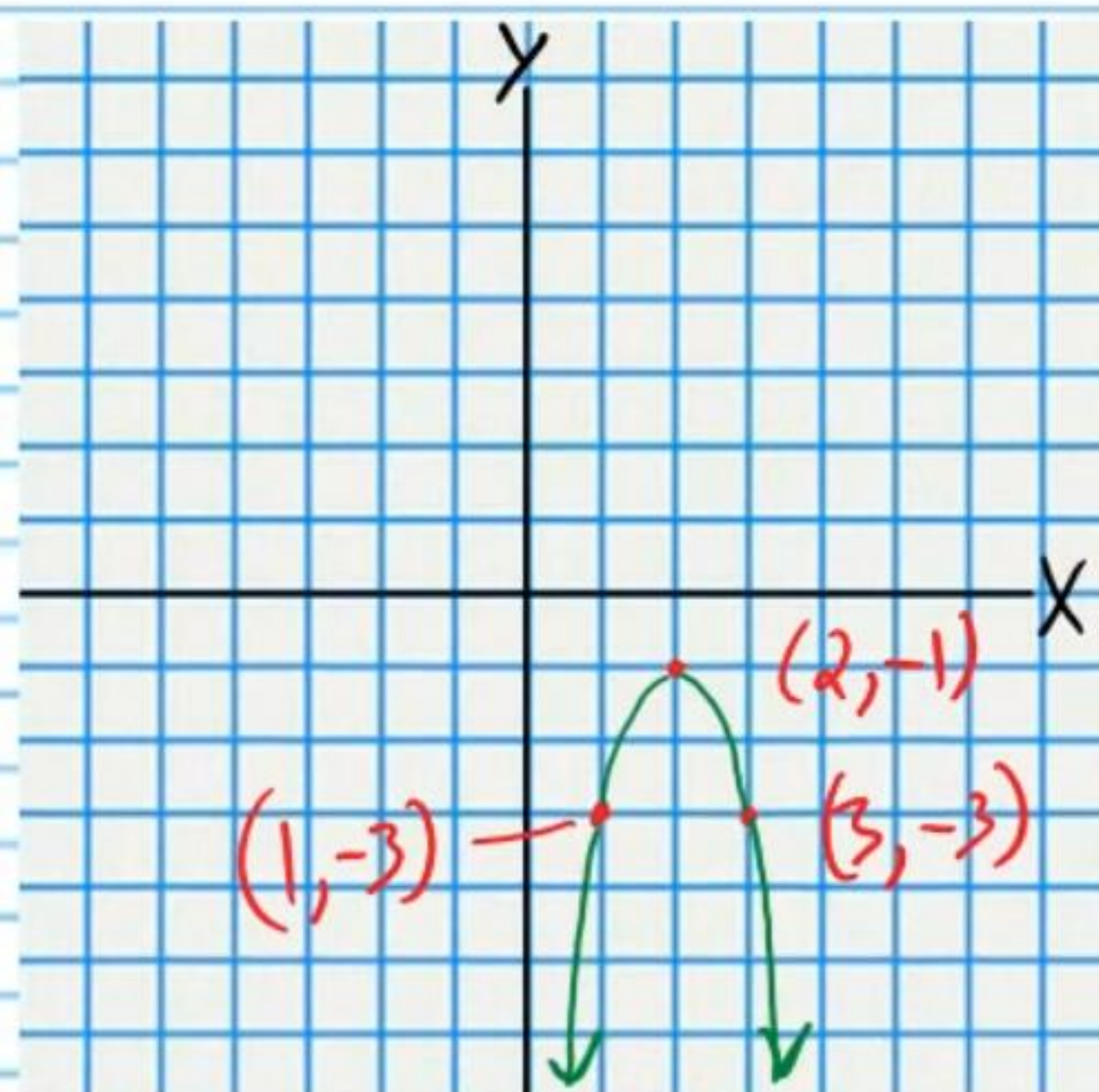
II) $x=1$

$y=-2(1-2)^2-1$

$y=-2(-1)^2-1$

$y=-2(1)-1$

$y=-2-1=-3$



$$(18) 3(x+4)^2 - 3 = y$$

$$a=3, h=-4, k=-3$$

$$I) (-4, -3)$$

$$II) x = -5$$

$$y = 3(-5+4)^2 - 3$$

$$y = 3(-1)^2 - 3$$

$$y = 3(1) - 3$$

$$y = 3 - 3$$

$$y = 0$$

x	y
-4	-3
-5	0
-3	0

